SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

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COURSE NAME: Computer Networks
COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an 8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.**

Questions

1. **Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer**.

The problem is mainly caused by the UDP transport protocol under poor network conditions and the application-layer buffering strategy.

- 8% packet loss causes missing voice/video packets.
- 180 ms jitter results in irregular packet arrival.
- Fluctuating bandwidth affects smooth video streaming.
- Insufficient application buffering worsens playback quality.

Hence, both UDP network behavior and application buffering contribute to the issue.

2. **Justify your answer based on the communication requirements of real-time multimedia applications.**

Real-time applications require:

- Low latency
- Low jitter
- Continuous media playback
- Fast packet delivery

Most conferencing applications use UDP because:

- No retransmission delays
- Faster communication
- Suitable for live voice and video

Although some packets may be lost, delayed packets are more harmful than lost packets in real-time communication.

3. **Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.**

Protocol-Level Improvements

- Use RTP over UDP.
- Enable Forward Error Correction (FEC).
- Implement QoS prioritization.
- Use adaptive bitrate streaming.

Application-Level Improvements

- Increase adaptive jitter buffer.
- Dynamic video quality adjustment.
- Packet loss concealment.
- Congestion control.

Result: Better voice quality, fewer frozen frames, and reduced delay.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.**

Questions

1. **Analyze the reasons behind the high DNS resolution time.**

   The DNS lookup time of **800 ms** is mainly caused by a centralized DNS infrastructure. Possible reasons include:

- A single DNS server handling all requests.
- Long geographical distance between users and the DNS server.
- High server workload during peak hours.
- Multiple recursive DNS queries.
- Poor DNS caching.
- Improper TTL configuration causing repeated lookups.

   These issues increase the time required to resolve domain names before loading the web application.

2. **Propose a suitable DNS architecture using techniques such as** Anycast DNS**,** DNS pre-fetching**, and** TTL optimization **to reduce the lookup time below** 50 ms**.**

   A **distributed DNS architecture** should be implemented to improve performance and ensure high availability.

   Proposed Architecture

Users Worldwide

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Local DNS Resolver

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▼ ▼ ▼

Anycast DNS   Anycast DNS   Anycast DNS

(Asia)   (Europe)   (America)

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Authoritative DNS Servers

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Web Application Servers

a) Anycast DNS

- Deploy multiple DNS servers across different continents.
- All servers share the same IP address.
- Internet routing automatically directs users to the nearest DNS server.
- This significantly reduces network latency and improves response time.

b) DNS Pre-fetching

- Frequently accessed domain names are resolved before the user requests them.
- Browsers or DNS resolvers store these records in advance.
- This reduces waiting time when users access the application.

c) TTL Optimization

- Configure TTL values to balance caching and record updates.
- Moderate TTL values (for example, 300–600 seconds) reduce repeated DNS lookups while still allowing timely updates when server addresses change.

d) DNS Caching

- Enable caching at browsers, operating systems, and local recursive DNS resolvers.
- Cached records eliminate unnecessary DNS queries, reducing lookup time.

e) Load Balancing

- Distribute DNS requests evenly among multiple servers.
- Prevents server overload and improves scalability.

f) Secondary DNS Servers

- Configure backup authoritative DNS servers.
- If one DNS server fails, another server automatically responds without interrupting service.

3. **Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.**

Aspect	Advantages	Limitations
Performance	Faster DNS resolution due to nearest Anycast server and local caching. Reduces lookup time below 50 ms.	Initial deployment and optimization require careful planning.
Scalability	Multiple DNS servers can easily support increasing numbers of global users without performance degradation.	Additional infrastructure increases management complexity.
Availability	Multiple DNS servers and automatic failover ensure uninterrupted service if one server fails.	Synchronizing DNS records across all servers requires proper management.
Reliability	Load balancing prevents server overload and improves system stability.	Incorrect TTL settings may delay updates or increase DNS traffic.
Network Efficiency	DNS caching and pre-fetching reduce unnecessary DNS queries and lower network traffic.	Cache consistency must be maintained when DNS records change.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly 800 ms **because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.**

Questions

1. Examine the impact of centralized DNS architecture on application performance.

A **centralized DNS architecture** means that all DNS queries are handled by one primary DNS server. While this design is simple to manage, it creates several performance and reliability issues when serving users worldwide.

Impact on Application Performance

a) High DNS Lookup Latency

Since users from different countries must contact the same DNS server, network propagation delay increases. This results in DNS lookup times of around **800 ms**, slowing down webpage loading.

b) Server Overload

During promotional events, millions of DNS requests reach the same server simultaneously. The server becomes overloaded, increasing response time and reducing overall performance.

c) Single Point of Failure

If the primary DNS server fails due to hardware problems, software issues, or network outages, users cannot resolve domain names, making cloud applications inaccessible.

d) Poor User Experience

Slow DNS resolution delays the establishment of HTTP/HTTPS connections. Users experience slow page loads, delayed application responses, and possible request timeouts.

e) Limited Scalability

A centralized DNS server cannot efficiently handle increasing global traffic. As the number of users grows, performance degrades further unless expensive hardware upgrades are made.

f) Increased Network Traffic

All DNS requests travel to a single location, consuming additional bandwidth and increasing network congestion.

2. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

To overcome the limitations of centralized DNS, the company should implement a **distributed DNS architecture**.

Proposed Architecture

Users Worldwide

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Local Recursive DNS

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Anycast DNS Anycast DNS Anycast DNS

(Asia) (Europe) (America)

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Authoritative DNS Servers

(Primary + Secondary)

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Cloud Application Servers

Components of the Solution

a) Anycast DNS

- Deploy DNS servers in multiple geographical regions.
- All servers share the same IP address.
- Internet routing automatically directs users to the nearest available DNS server.
- This reduces lookup latency significantly.

b) Multiple Authoritative DNS Servers

- Configure primary and secondary authoritative DNS servers.
- If one server fails, another immediately handles DNS requests.
- Ensures continuous service availability.

c) DNS Caching

- Enable caching at browsers, operating systems, and recursive DNS resolvers.
- Frequently accessed records are served from cache, reducing lookup time and server load.

d) Load Balancing

- Distribute DNS traffic evenly across multiple servers.
- Prevents server overload during traffic spikes.

e) Health Monitoring and Automatic Failover

- Continuously monitor DNS server health.
- Automatically redirect traffic if a server becomes unavailable.

Expected Outcome

- DNS lookup time reduced from **800 ms to below 50 ms**.
- Faster application access.
- Better scalability during peak traffic.
- High availability with no single point of failure.

3. Analyze how TTL values**,** Anycast routing**, and** DNS caching **influence system performance and fault tolerance.**

a) TTL (Time-To-Live) Values

TTL specifies how long a DNS record remains in cache before it must be refreshed.

Influence on Performance

- Higher TTL values reduce repeated DNS queries.
- Improves response time because cached records are reused.
- Reduces workload on authoritative DNS servers.

Influence on Fault Tolerance

- Lower TTL values allow DNS changes to propagate quickly during server failures.
- Extremely low TTL values increase DNS traffic and server load.
- A balanced TTL (e.g., **300–600 seconds**) provides both good performance and flexibility.

b) Anycast Routing

Anycast allows multiple DNS servers in different locations to share the same IP address.

Influence on Performance

- Routes users to the nearest DNS server.
- Reduces network latency and lookup time.
- Improves response speed for global users.

Influence on Fault Tolerance

- If one DNS server becomes unavailable, routing automatically redirects users to another healthy server.
- Eliminates the single point of failure.
- Improves overall service reliability.

c) DNS Caching

DNS caching stores previously resolved DNS records at browsers, operating systems, and recursive DNS servers.

Influence on Performance

- Eliminates repeated DNS lookups.
- Reduces lookup time to a few milliseconds for cached records.
- Decreases network traffic and DNS server workload.

Influence on Fault Tolerance

- Cached records continue to be used temporarily even if the authoritative DNS server becomes unavailable.
- Excessively long cache durations may delay the propagation of updated DNS records.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the average order acknowledgment latency increases from 1 ms to 40 ms, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. **Analyze three possible TCP-related factors responsible for the latency increase, considering flow control**, **Nagle's algorithm**, and **SYN queue limitations**.

The increase in latency is mainly due to three TCP-related factors:

a) TCP Flow Control

TCP flow control ensures that the sender does not transmit data faster than the receiver can process it. It uses the **TCP receive window (Sliding Window mechanism)** to control the amount of unacknowledged data in transit.

Analysis

- During peak trading hours, the receiver's buffer fills quickly due to the large number of incoming orders.
- TCP reduces the sending rate to prevent buffer overflow.
- The sender must wait for acknowledgments before sending additional data.
- This increases order acknowledgment latency and slows communication.

b) Nagle's Algorithm

Nagle's Algorithm combines multiple small TCP packets into a larger packet before transmission to reduce network congestion.

Analysis

- Stock trading systems generate many **small order messages**.

- Instead of sending each order immediately, Nagle's Algorithm waits for previous packets to be acknowledged.
- This introduces additional delay before the order is transmitted.
- Although network efficiency improves, response time increases, which is unacceptable in high-frequency trading.

c) SYN Queue Limitation

Before a TCP connection is established, incoming connection requests are temporarily stored in the server's **SYN queue**.

Analysis

- During market opening, thousands of traders connect simultaneously.
- The SYN queue becomes full due to excessive connection requests.
- New connection requests are delayed or dropped.
- Longer connection establishment times increase overall transaction latency.

2. **Explain how each factor affects transaction processing in high-frequency systems**

In electronic stock trading, every millisecond is important because delayed transactions may result in missed trading opportunities and financial losses.

a) Effect of Flow Control

- Limits the rate at which trading orders are transmitted.
- Delays acknowledgments when receiver buffers become full.
- Slows order execution during heavy trading periods.
- Reduces overall system throughput.

Impact: Slower trade confirmations and delayed execution of buy/sell orders.

b) Effect of Nagle's Algorithm

- Delays transmission of small TCP packets.
- Waits for acknowledgment before sending additional packets.
- Increases communication delay for individual trade requests.

Impact: Even a few milliseconds of extra delay may affect stock prices and reduce trading efficiency.

c) Effect of SYN Queue Limitation

- Delays establishment of new TCP connections.
- Some client requests may timeout or be dropped.
- Users experience slow login or delayed order submission during peak traffic.

Impact: Increased customer dissatisfaction and reduced reliability of the trading platform.

3. **Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.**

a) Disable Nagle's Algorithm (TCP_NODELAY)

- Configure **TCP_NODELAY** to send each trading order immediately without waiting for additional packets.
- Reduces transmission delay for small messages.

Benefit: Faster order execution and lower latency.

b) Increase TCP Send and Receive Buffer Sizes

- Increase socket buffer sizes to handle high traffic volumes.
- Prevent receiver buffer overflow and reduce flow-control delays.

Benefit: Higher throughput and smoother data transfer.

c) Increase SYN Backlog Queue Size

- Increase the maximum number of pending TCP connection requests.
- Allows the server to handle connection bursts during market opening.

Benefit: Faster connection establishment and fewer dropped requests.

d) Enable SYN Cookies

- Protects the server from SYN queue overflow.
- Allows the server to continue accepting legitimate connections during heavy traffic.

Benefit: Improved reliability and protection against connection floods.

e) Enable TCP Fast Open (TFO)

- Reduces the TCP three-way handshake delay by allowing data to be sent during connection establishment.

Benefit: Faster client connections and reduced startup latency.

f) Use Persistent TCP Connections

- Reuse existing TCP connections instead of creating a new connection for every order.

Benefit: Eliminates repeated connection setup delays and improves response time.

g) Optimize TCP Congestion Control

- Use modern congestion control algorithms such as **BBR** or **CUBIC** based on the network environment.
- Adjust retransmission timeout (RTO) parameters where appropriate.

Benefit: Reduces unnecessary retransmissions and improves network utilization.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand